



IMAGE Information Systems Ltd.

For a Better View in Diagnostics!

iQ-DICOMTEST

User Manual

v 3.1

GLOSSARY

AE	Application Entity
AET	Application Entity Title
ASCII	American Standard Code for Information Interchange
DICOM	Digital Imaging and Communications in Medicine
IP	Internet Protocol
JPEG	Joint Photographic Experts Group
SCP	Service Class Provider
SCU	Service Class User

Table of Contents

0	DISCLAIMER	4
1	INTRODUCTION	5
2	PLATFORMS	6
3	INSTALLATION	6
4	SETUP	6
4.1	Network Setup.....	6
4.1.1	Log Level	7
4.2	Printer Setup	9

0 Disclaimer

IMAGE Information Systems Ltd. took the utmost care in developing iQ-DICOMTEST. Completed tests on a variety of different computer systems did not show any indication of errors. In spite of passing a series of tests we can not completely guarantee for an error-free execution of iQ-DICOMTEST on your target system.

Under no circumstances IMAGE Information Systems can be held liable for damage of any kind resulting from the use of our product, regardless of whether it is direct or indirect damage, resultant damage or specific damage including profit loss or damage resulting from the loss of data. This documentation does not assert claims to accuracy and integrity.

iQ-DICOMTEST is not a medical device. It is not allowed to use for any medical purposes except technical testing.

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1 Introduction



IQ-DICOMTEST represents a set of tools for the purpose of testing and troubleshooting DICOM communications. The software acts as a DICOM SCU (**S**ervice **C**lass **U**ser) and is able to perform the following services:

Echo – Verification SCU	to test the basic communication with a SCP (Service Class Provider)
Storage SCU	sends a DICOM object using different transfer syntaxes
Query – Find SCU	queries a remote database using “patient name” or “study data”
Worklist SCU	able to query a remote Worklist SCP using “patient name” or “study data”
DICOM Print SCU	tests the printing capability of DICOM, JPEG or Bitmap images on a remote DICOM printer for grayscale as well as color output
Dump	shows the content of a DICOM object

2 Platforms

The following platforms are currently supported:

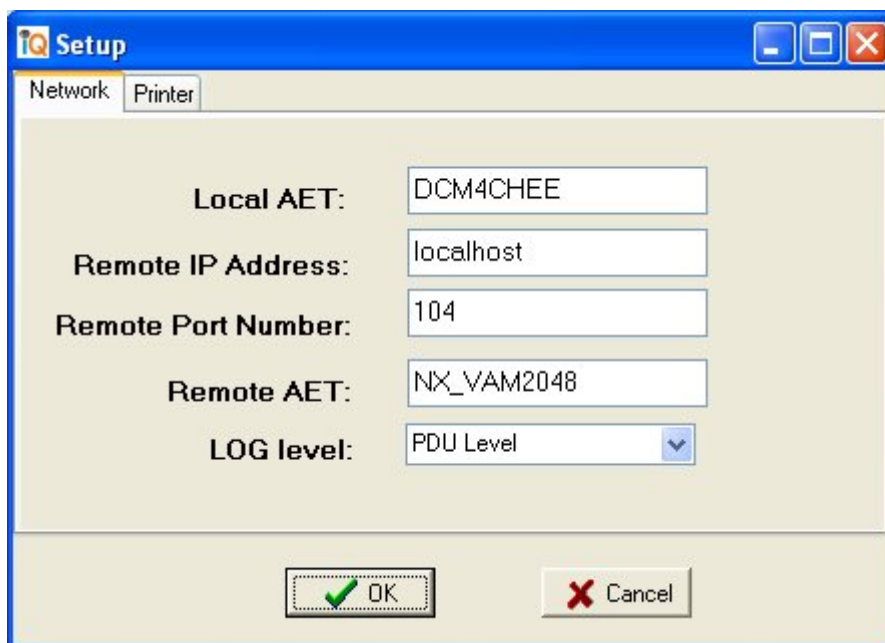
- Windows 2000 Server & Workstation
- Windows XP Professional
- Windows Server 2003
- Windows Vista Home Basic/Home Premium/Business/Enterprise/Ultimate

3 Installation

IQ-DICOMTEST can be executed without any system-installation.

4 Setup

4.1 Network Setup



The screenshot shows a Windows-style dialog box titled "IQ Setup". It has two tabs: "Network" (selected) and "Printer". The "Network" tab contains five input fields with labels to their left: "Local AET:" with the value "DCM4CHEE", "Remote IP Address:" with the value "localhost", "Remote Port Number:" with the value "104", "Remote AET:" with the value "NX_VAM2048", and "LOG level:" with a dropdown menu showing "PDU Level". At the bottom of the dialog are two buttons: "OK" (with a green checkmark icon) and "Cancel" (with a red X icon).

The following parameters can be changed in this dialog window:

Local AET	iQ-DICOMTEST Application Entity Title
Remote IP Address	IP-Address of the remote device
Remote Port Number	port number for the remote device
Remote AET	remote device Application Entity Title
LOG Level	represents the different logging levels (No Log, PDU Level, or Socket Level)

4.1.1 Log Level

It is possible to choose between the three following log structures:

No Log

In this mode no information about the DICOM communication is recorded.

PDU Level

By selecting the option “PDU Level” DICOM communication is logged on Tag Level. Each tag contains additional information in form of “(group, element), length, VR, “value” – Description”.

Have a look on the following example of an ECHO communication for further details.

```
10:15:48 AM: Starting ECHO Service
Connected to: 192.168.2.4:11112
Presentation Context Accept, Transfer Syntax: 1.2.840.10008.1.2
DCM4CHEE >> NX_VAM2048
    (0000,0000), 4, UL, "56" - Group Length
    (0000,0002), 18, UI, "1.2.840.10008.1.1" - Affected SOP
Class UID
    (0000,0100), 2, US, "48" - Command Field
    (0000,0110), 2, US, "3" - Message ID
    (0000,0800), 2, US, "257" - Data Set Type
DCM4CHEE << NX_VAM2048
    (0000,0002), 2, UI, "1." - Affected SOP Class UID
    (0000,0100), 2, US, "32816" - Command Field
    (0000,0120), 2, US, "3" - Message ID Being Responded To
    (0000,0800), 2, US, "257" - Data Set Type
    (0000,0900), 2, US, "0" - Status
10:15:48 AM: ECHO OK
```

Socket Level

By clicking on “Socket Level” all Bytes that have passed the socket are shown in the log file in both hexadecimal and ASCII format. Length and direction of each byte package can be found at the start.

```
10:03:29 AM: Starting ECHO Service
Connected to: 192.168.2.4:11112
74 >> 01 00 00 00 00 CD 00 01 00 00 4E 58 5F 56 41 4D 32 30 34
38 00 00 00 00 00 00 44 43 4D 34 43 48 45 45 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
.....NX_VAM2048.....DCM4CHEE.....
.....
25 >> 10 00 00 15 31 2E 32 2E 38 34 30 2E 31 30 30 30 38 2E 33
2E 31 2E 31 2E 31
....1.2.840.10008.3.1.1.1
8 >> 20 00 00 2E 01 00 00 00
.....
21 >> 30 00 00 11 31 2E 32 2E 38 34 30 2E 31 30 30 30 38 2E 31
2E 31
0...1.2.840.10008.1.1
```

```

21 >> 40 00 00 11 31 2E 32 2E 38 34 30 2E 31 30 30 30 38 2E 31
2E 32
@...1.2.840.10008.1.2

4 >> 50 00 00 3A
P...

8 >> 51 00 00 04 00 00 40 00
Q.....@.

34 >> 52 00 00 1E 31 2E 32 2E 38 32 36 2E 30 2E 31 2E 33 36 38
30 30 34 33 2E 32 2E 31 33 39 36 2E 39 39 39
R...1.2.826.0.1.3680043.2.1396.999

16 >> 55 00 00 0C 43 68 61 72 72 75 61 56 69 73 74 61
U...CharruaVista

74 << 02 00 00 00 00 C0 00 01 00 00 4E 58 5F 56 41 4D 32 30 34
38 00 00 00 00 00 20 44 43 4D 34 43 48 45 45 00 00 00 00 00 00
20 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
.....NX_VAM2048..... DCM4CHEE.....
.....

124 << 10 00 00 15 31 2E 32 2E 38 34 30 2E 31 30 30 30 38 2E 33
2E 31 2E 31 2E 31 21 00 00 19 01 00 00 00 40 00 00 11 31 2E 32 2E
38 34 30 2E 31 30 30 30 38 2E 31 2E 32 50 00 00 42 51 00 00 04 00
00 40 00 52 00 00 1E 31 2E 32 2E 38 32 36 2E 30 2E 31 2E 33 36 38
30 30 34 33 2E 32 2E 31 33 39 36 2E 39 39 39 55 00 00 14 43 68 61
72 72 75 61 53 6F 66 74 5F 50 72 69 6E 74 53 43 50
....1.2.840.10008.3.1.1.1!.....@...1.2.840.10008.1.2P..BQ.....@.
R...1.2.826.0.1.3680043.2.1396.999U...CharruaSoft_PrintSCP

Presentation Context Accept, Transfer Syntax: 1.2.840.10008.1.2
DCM4CHEE >> NX_VAM2048
(0000,0000), 4, UL, "56" - Group Length
(0000,0002), 18, UI, "1.2.840.10008.1.1" - Affected SOP
Class UID
(0000,0100), 2, US, "48" - Command Field
(0000,0110), 2, US, "3" - Message ID
(0000,0800), 2, US, "257" - Data Set Type

80 >> 04 00 00 00 00 4A 00 00 00 46 01 03 00 00 00 00 04 00 00
00 38 00 00 00 00 02 00 12 00 00 00 31 2E 32 2E 38 34 30 2E 31
30 30 30 38 2E 31 2E 31 00 00 00 00 01 02 00 00 00 30 00 00 00 10
01 02 00 00 00 03 00 00 00 00 08 02 00 00 00 01 01
.....J...F.....8.....1.2.840.10008.1.1.....0.....
.....

62 << 04 00 00 00 00 38 00 00 00 34 01 03 00 00 02 00 02 00 00
00 31 2E 00 00 00 01 02 00 00 00 30 80 00 00 20 01 02 00 00 00 03
00 00 00 00 08 02 00 00 00 01 01 00 00 00 09 02 00 00 00 00 00
.....8...4.....1.....0.....

DCM4CHEE << NX_VAM2048
(0000,0002), 2, UI, "1." - Affected SOP Class UID
(0000,0100), 2, US, "32816" - Command Field
(0000,0120), 2, US, "3" - Message ID Being Responded To
(0000,0800), 2, US, "257" - Data Set Type
(0000,0900), 2, US, "0" - Status

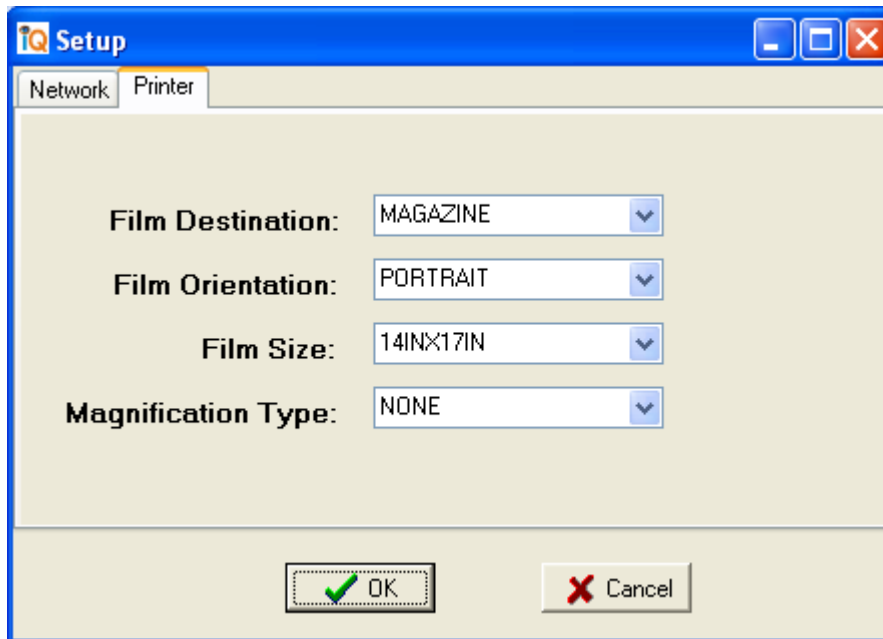
10 >> 05 00 00 00 00 04 00 00 00 00
.....

10 << 06 00 00 00 00 04 00 00 00 00
.....

10:03:29 AM: ECHO OK

```

4.2 Printer Setup



This works exclusively for the DICOM Print SCU service.

Film Destination	MAGAZINE, PROCESSOR, BIN_1, BIN_2 or BIN_3
Film Orientation	PORTRAIT or LANDSCAPE
Film Size	14INX17IN, 14INX14IN, 10INX12IN, 8INX10IN
Magnification Type	REPLICATE, BILINEAR, CUBIC or NONE